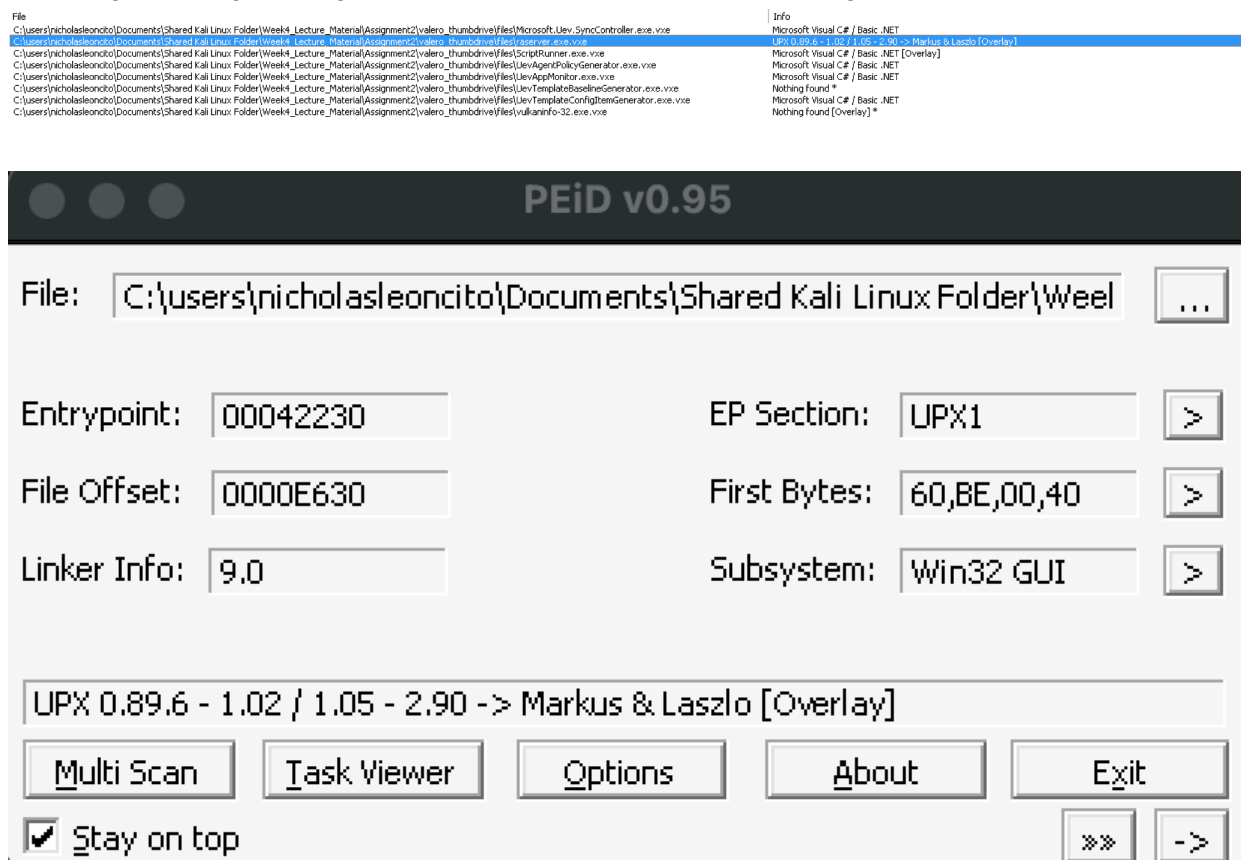


## Assignment 2

Questions to answer:

1. Which of the executable files is the OnyxCrew malware? Submit a screenshot supporting your findings.

The executable file OnyxCrew used as malware is raaserver.exe. I first used PEiD to scan the folder containing all the exe files on the thumbdrive. UPX was listed in the info tab for Raaserver this was also shown as a red flag for possible malware, due to malware usually being packed to not being able to get strings out of a packet as well as obfuscating the codebase.



2. Once unpacked, what is Autorun persistence registry key embedded within the malware? (Submit a screenshot supporting your findings)

After unpacking, I found the malware had an autorun persistence key. This allows the program to launch every time the system boots.

```
PS C:\bstrings> ./bstrings -f "Z:\Week4_Lecture_Material\Assignment2\valero_thum
bdrive\files\raserver_unpacked.exe.vxe" --lr reg_path
bstrings version 1.5.1.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/bstrings

Command line: -f Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\r
aserver_unpacked.exe.vxe --lr reg_path

Searching via RegEx pattern: ([a-z0-9]\\)*(software\\)|(sam\\)|(system\\)|(secur
ity\\)[a-z0-9\\]+

Searching 1 chunk (512 MB each) across 249.867 KB in 'Z:\Week4_Lecture_Material\
Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe'
Chunk 1 of 1 finished. Total strings so far: 3,826 Elapsed time: 0.214 seconds.
Average strings/sec: 17,822
Primary search complete. Looking for strings across chunk boundaries...
Search complete.

Processing strings...

HKL \Software\Microsoft\Windows\CurrentVersion\Run\raserver.exe

Found 1 string in 0.217 seconds. Average strings/sec: 17,632
PS C:\bstrings> ./bstrings -f "Z:\Week4_Lecture_Material\Assignment2\valero_thum
bdrive\files\raserver_unpacked.exe.vxe" --lr reg_path -q
HKL \Software\Microsoft\Windows\CurrentVersion\Run\raserver.exe
PS C:\bstrings> |
```

3. What is the windows file path embedded within the malware? (Submit a screenshot supporting your findings)

After unpacking, I found the malware had an autorun persistence key. This allows the program to launch every time the system boots.

The windows file path embedded within the malware is c:\OnyxCrew\ransware\raserver\release

```

PS C:\bstrings> ./bstrings -f "Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe" --lr win_path -q
c:\OnyxCrew\ransomware\raserver\Release\
PS C:\bstrings> ./bstrings -f "Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe" --lr win_path -
bstrings version 1.5.1.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/bstrings

Command line: -f Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe --lr win_path-

Searching via RegEx pattern: (?:"?[a-zA-Z]\:|\\\\[^\\"/\\:\*\?<>\\|]+\\[^\\"/\\:\*\?<>\\|]*)\\(?:[^\\"/\\:\*\?<>\\|]+\\)*.*w([^\\"/\\:\*\?<>\\|])*

Searching 1 chunk (512 MB each) across 249.867 KB in 'Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe'

Chunk 1 of 1 finished. Total strings so far: 3,826 Elapsed time: 0.117 seconds.
Average strings/sec: 32,587
Primary search complete. Looking for strings across chunk boundaries...
Search complete.

Processing strings...

c:\OnyxCrew\ransomware\raserver\Release\

Found 1 string in 0.119 seconds. Average strings/sec: 32,049
PS C:\bstrings> |

```

4. What is the URL embedded within the malware? (Submit a screenshot supporting your findings)

I used the bstrings command <file path> -lr url3986 to search for any Urls. This can be something important like the malware sending traffic to a domain or even being tied to the malware.

The Url takes you to a QR code that when scanned provides a bitcoin address.

```
PS C:\bstrings> ./bstrings -f "Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe" --lr url3986
bstrings version 1.5.1.0
```

```
Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/bstrings
```

```
Command line: -f Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe --lr url3986
```

```
Searching via RegEx pattern: ^
```

|                                              |                  |
|----------------------------------------------|------------------|
| [a-z][a-z0-9+\-.]*://                        | # Scheme         |
| ([a-z0-9\-.~%!\\$&'()*+,\;=:@]?)             | # User           |
| (?<host>[a-z0-9\-.~%]+                       | # Named host     |
| \[a-f0-9:]+\]                                | # IPv6 host      |
| \[v[a-f0-9][a-z0-9\-.~%!\\$&'()*+,\;=:@]+\]) | # IPvFuture host |
| (:[0-9]+)?                                   | # Port           |
| (/[a-z0-9\-.~%!\\$&'()*+,\;=:@]+)*/?         | # Path           |
| (\[a-z0-9\-.~%!\\$&'()*+,\;=:@/?]*)?         | # Query          |
| (\[a-z0-9\-.~%!\\$&'()*+,\;=:@/?]*)?         | # Fragment       |
| \$                                           |                  |

```
Searching 1 chunk (512 MB each) across 249.867 KB in 'Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe'
```

```
Chunk 1 of 1 finished. Total strings so far: 3,826 Elapsed time: 0.175 seconds.
Average strings/sec: 21,899
Primary search complete. Looking for strings across chunk boundaries...
Search complete.
```

```
Processing strings...
```

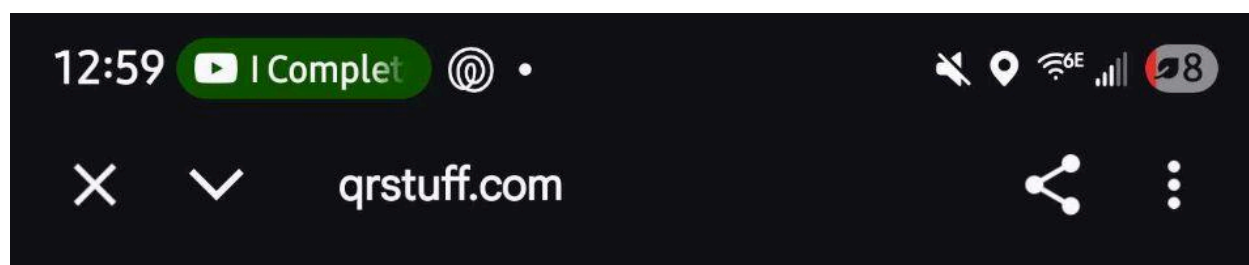
```
https://tinyurl.com/2p865z29
```

```
Found 1 string in 0.176 seconds. Average strings/sec: 21,689
```

```
PS C:\bstrings> ./bstrings -f "Z:\Week4_Lecture_Material\Assignment2\valero_thumbdrive\files\raserver_unpacked.exe.vxe" --lr url3986 -q
```

```
https://tinyurl.com/2p865z29
```

```
PS C:\bstrings>
```



# QR Code Scanner & Reader Online

The QRStuff QR Code Scanner is the perfect free tool to scan QR code online, anytime and anywhere. Now you can easily and quickly generate, download, and scan your QR codes all in one place.

## Scan Result

bc1q7cyrfmck2ffu2ud3rn5l5a8yv6f0chkp0zpemf

New scan

5. After visiting the URL, what is the Bitcoin Wallet address embedded at the website? (Submit a screenshot supporting your findings)

At the time of looking this up at 1am this is the conversion 978.23242699 BTC = \$113,249,956.41

The screenshot shows a web interface for looking up a Bitcoin address. At the top, the heading "Bitcoin address lookup" is displayed in a large, dark font. Below it, a subtitle reads "Enter a public address to check a Bitcoin wallet balance". A search bar contains the address "bc1q7cyrfmck2ffu2ud3rn5l5a8yv6f0chkp0zpemfj" with a clear button (X) to its right. To the right of the search bar is a dropdown menu showing "BTC" with a Bitcoin icon and a downward arrow. A "Check balance" button is positioned to the right of the dropdown. Below the search bar, a box displays the "Bitcoin wallet balance" as "978.23242699 BTC" followed by the dollar amount "\$113,249,956.41". Below this, the timestamp "Sep 20, 2025 1:02 AM" is shown. At the bottom of the box is a button labeled "Add wallet to CoinTracker".

# Bitcoin address lookup

Enter a public address to check a Bitcoin wallet balance

Q bc1q7cyrfmck2ffu2ud3rn5l5a8yv6f0chkp0zpemfj (X) BTC ▼ Check balance

Bitcoin wallet balance

**978.23242699 BTC** \$113,249,956.41

Sep 20, 2025 1:02 AM

Add wallet to CoinTracker